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Senior Design

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Individual Progress Memo 2

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Introduction

This capstone is a sponsored project by Lakeland Regional Health Center at Florida Polytechnical University. Our objective is to add AI chatbot functionality to their existing doctor search, so that patients can search what their symptom is and get the best results in their area. As a team, we have specific goals including developing and integrating a professional chatbot user interface, which interacts with the sponsor's infrastructure, as well as creating a text embedding model tailored to their business.

Progress Made Since Progress Memo 1

- Created a SQLite schema matching the spreadsheet that the sponsors gave the team. With it I made a python script to fill the database with dummy data. Later it will be used to show prototypes without leaking actual physicians.
- Created a prototype for the demonstration, for the backend I used python using langchain to interact with chatGPT, and allow the LLM to interact with the database through a query builder. The front end chat bot was quickly built using a library called Gradio. It was a simple setup to show proof of concept.

- Modified the custom chat bot user interface that Preston made, adding some ui changes and logic to be able to send and receive prompt requests. Using python flask, I was able to host the backend of the chatbot.
- Made a high level diagram of the project, in addition made video presentations of the prototype and explained it on the presentation.

Plan for Completion and Showcase

Similar tasks I have done:

This semester I am taking artificial intelligence, which taught prompt engineering and how to build complex systems integrating AI, using langchain. This helped enormously as it allowed me to create a basic prototype.

In addition I have extensive knowledge in python, JS, and SQLite. All three technologies I have used to create several projects. It allowed me to quickly build out a system.

Core tasks:

As of April 1, we have successfully trained the text embedding model called BERT.

The next task is to use and interact with the mode using python. The current workflow will look like. The backend/frontend and its input/outputs are already built in the prototype. All we have to do is to integrate it with the prototype. Everyone in the group will work on this as time is running out.

	<- user prompt		<- take prompt	
BERT		backend		User interface
	SQL query ->		shows the result of the SQL query ->	

Another task is to build the final presentation and report. Changing from our last demonstration we will go greatly in depth in how the system works and better explain what the model is and how we trained it.

The last task is to build a polished version of what we already have, based on our next meeting with the sponsors. This may require integrating the prototype to fit with their servers and modifying the UI in their style.